

ANITA CAROLYNE A. ORERA

GIS Analyst / Data Specialist

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With a solid background in Geomatics Engineering and GIS encompassing aspects of Data analysis, Cartography, Remote Sensing, Digital Mapping, and ETL processes, I am an Innovative Data & GIS Analyst with over 3 years of experience driving impactful solutions in automated data scraping, geospatial analysis, and AI data services. I am proficient in building structured data extraction workflows to ensure high-fidelity data normalization. I am passionate about transforming data into actionable insights, from online data collection to final visualization and presentation, ultimately supporting decision-making and strategic organizational objectives.

Work Experience

Geospatial Analyst

Feb 2025 – Present

Ecospace Services Ltd | Hybrid, Nairobi

I apply my expertise in geospatial data analysis, GIS technologies, and spatial modelling to help deliver intelligent, data-driven solutions to a variety of sectors. This is a multi-faceted role that blends technical expertise with instructional leadership. Among other responsibilities, I actively engage in;

Geospatial Analysis & Data Management

- **End-to-End Data Processing:** Gathering and processing multi-source geospatial data (*Satellite imagery, GPS, and field surveys*) using ArcGIS, QGIS, R and Google Earth Engine.
- **Spatial Modelling:** Performing complex spatial analysis and remote sensing to provide insights into land use, urban growth, and environmental resource management through activities inclusive of computation of indices, time series analysis and charting for various tasks.
- **Database Administration:** Design of GIS databases using PostgreSQL, ensuring capturing of user needs, data consistency and accuracy for cross-functional project teams.
- **Data Extraction & Scraping:** Research, identify, and systematically extract data from public repositories, government portals, APIs, and online feeds using R and Python script automation.
- **Data Transformation & Normalization:** Expertly transforming, cleaning, and normalization of complex tabular and spatial data into standardized PostgreSQL and relational database schemas to ensure absolute consistency.
- **Visualization and Reporting:** Designing detailed maps, charts and advising on interactive geospatial dashboards, web portals to support data-driven decision-making for different use cases.
- **Collaboration & Project Support:** Supporting cross-functional teams, senior analysts and managers in the development of geospatial solutions and the presentation of findings to clients.

Training, Quality Assurance & Instructional Leadership

- **Program Development:** Authoring comprehensive training manuals, e-learning videos, and technical documentation for GIS and remote sensing software.
- **Quality Assurance & Documentation:** Refining data validation workflows to catch inconsistencies, audited data samples, and authoring of technical documentation/methodology manuals for all automated data collection processes.
- **Instructional Delivery:** Conducting virtual training for internal teams and external clients to facilitate the adoption of geospatial solutions.
- **Impact Assessment:** Developing performance metrics and post-training assessments to measure effectiveness and continuously improve training curriculum.

GIS, Remote Sensing Analyst - Research and Development

Aug 2023 - Feb 2024

Regional Centre for Mapping of Resources for Development (RCMRD) | Kasarani, Nairobi

Performing GIS data scraping, analysis, prediction and mapping under two main projects; - GMES & Africa (Global Monitoring for Environment and Security and Africa) - Geo-mountains small grants program 2023 (Land Cover Change Assessment & Invasive Species Mapping.)

- **Wetland monitoring and assessment:** a subsection which involved an initial research process to extract relevant data, creation of maps for visualization and generation of statistics for analysis. 90% of the listed Somalia wetlands were efficiently mapped and analyzed using Jupyter Notebooks (Python) showing wetland changes over the different time periods.
- **Land Degradation Assessment:** Correction of Uganda's NDVI product ensuring 100% accuracy in input data

- **Land Cover Change Detection:** Programming on Google Earth Engine, and integration with python programming was done to achieve the visualization of land cover changes around the Mount Kenya Region over a period of 30 years. Graphs and charts were similarly generated to aid in trend visualization statistics.
- **Invasive Species Mapping:** Data modelling using the R programming environment produced accuracies above 70% aiding in the visualization of predicted spread and associated factors.
- **Repository Management:** Maintained script repositories using Git version control, adhering to development best practices and collaborative team workflows.

UX Designer, Database Designer

Jan 2022 - Apr 2023

Ecospace Services Ltd | Remote, Nairobi

Engagement in the database design and UI/UX design processes

- Accurately modelled 90% of the entire working of the proposed database system using PostGIS and PostgreSQL.
- Successfully conducted research on user needs informing 80% of the preliminary workflow and structure of the apps involved.
- Working as the lead database designer, I enhanced and improved the database structure of the previously developed mobile and web applications under set timelines.

Geospatial Systems Attachee

Nov 2021 - Apr 2022

Jomo Kenyatta University of Agriculture and Technology (JKUAT) | Juja, Kiambu

Spearheaded the development of geospatial applications for a Small-scale crop-mapping project using machine learning approaches as the GIS student project lead.

- Successfully guided and coordinated the team to design and develop User Interface screens for the mobile and web application solving more than 60% of the listed gaps and problems faced.
- Formulated and worked on the development of the backend for the application
- Obtained and manipulated several spectral indices and the relevant values using Google Earth Engine and R programming language aside from handling communication between the associated parties.

Geospatial Systems Attachee

Feb 2021 - Apr 2021

Orbital Africa | Ngong Road, Kenya

Achieved simplification of data acquisition, automation of data collection, cleaning, analysis, evaluation and visualization to enhance the data mining process using R programming language.

- **Automated scraping:** Maximized geospatial data capture using ODK collect and performed online data retrieval through web scraping with R, fully automating what was previously a manual data-entry process.
- **Records Management:** Systematically maintained and structured information records from the automated web scraping processes.

Volunteer

Vice President, Treasurer, Secretary General

Jan 2019 - Dec 2021

LEAP JKUAT Chapter

Serving in three different roles as Vice President (2021), Treasurer (2020) and Secretary General (2019), I got to understand the workings of a leadership committee. I additionally familiarized myself with the need for structure and focused leadership.

Core Skills

Data Scraping & Extraction, R programming, ETL Pipelines, PostgreSQL/PostGIS, Data Normalization, Data Analysis, Git, Database Design, Google Earth Engine, Python, Machine Learning, Technical Writing, Documentation, Data Visualization, Microsoft Suite (Excel, Word, PowerPoint), GIS Software (ArcGIS/QGIS), Geographical Information Systems (GIS), UI-UX designing, Teamwork, Computer skills, Active Learning, Communication, AI Prompting, Data Visualization, Fast Learning, Remote Sensing, Spatial Modelling, Instructional Design, Technical Writing, E-learning Development, Data Visualization

Education

Jomo Kenyatta University of Agriculture and Technology

Bachelor of Science | Geomatics Engineering and Geospatial Information Systems

Second Class Honors (Upper Division)

Languages

English: fluent (*Native/Bilingual Proficiency*)

Kiswahili: fluent (*Native/Bilingual Proficiency*)